Figure 2.3 In this enterprise data warehouse socio-technical architecture, the boxes represent areas of responsibilities of different teams, the "ETL or ELT" and "ETL" rectangles are data transformation components, and the arrows indicate direction of call—from caller to callee.

In the example data warehouse in figure 2.3, you can see that teams A, B, and C are producing data and do not own the pipelines used to extract data from their sources. A central data team owns the pipelines. It is expected that the central team also owns BI and reporting. BI/reporting and data analysis in figure 2.3 are depicted as being owned by separate teams. But that's often not the case, and they also may be owned by the same central data team.

The following illustrates when you should consider implementing an enterprise data warehouse instead of a data mesh:

- Most of the sources of the data are structured.
- Your current data use cases are well known and not likely to expand or evolve.
- You extract value from data using mainly BI, reporting, and data analysis.
- You rely on a small number of unchanging source systems.
- Sources are owned by a small number of teams.
- Your central data team has expertise in data warehouse technologies.

As we stated in the beginning, the enterprise data warehouse is a socio-technical approach, but a data warehouse is an underlying technology. The technology itself is not an alternative to a data mesh. However, it might be one of the technological solutions used to build a data mesh. But remember, in such a case, a data warehouse should not try to aggregate data from the whole organization. A data warehouse's usefulness as a technological data storage solution should be decided by each domain on a case-by-case basis.

The following shows when you should consider using a data warehouse in addition to a data mesh: